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Batch 15

Algorithm

1. Start.
2. Read the size of the array, `size`.
3. If `size <= 0`, display `"Invalid size"` and stop.
4. Declare a variable-length integer array `arr[size]`.
5. Read all elements into `arr[]`.
6. Pass the array to `next_greater()` using its address.
7. Inside `next_greater()`:
   - Use an outer loop to select each element.
   - Initially assume its next greater element is `-1`.
   - Use an inner loop starting from the element immediately to its right.
   - Compare each right-side element with the current element.
   - When the **first greater element** is found, store it and `break` the inner loop. This ensures the nearest greater element is selected.
   - Display the result.
8. Separate results using commas.
9. Stop.

C Program

```
#include <stdio.h>

void next_greater(int *arr, int size)
{
    int i, j, next;

    for (i = 0; i < size; i++)
    {
        next = -1;

        /* Search elements to the right */
        for (j = i + 1; j < size; j++)
        {
            if (*(arr + j) > *(arr + i))
            {
                next = *(arr + j);
                break;
```

```
            }
        }

        printf("%d", next);

        if (i < size - 1)
        {
            printf(", ");
        }
    }

    printf("\n");
}

int main()
{
    int size, i;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid size.\n");
        return 0;
    }

    /* Variable-length array */
    int arr[size];

    printf("Enter %d elements: ", size);

    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    /* Pass the array using its address */
    next_greater(arr, size);

    return 0;
}
```

Test Case Table

Test	Size	Input Array	Expected Output
1	4	4 5 2 25	5, 25, 25, -1
2	4	13 7 6 12	-1, 12, 12, -1
3	4	1 3 2 4	3, 4, 4, -1
4	5	1 2 3 4 5	2, 3, 4, 5, -1
5	5	5 4 3 2 1	-1, -1, -1, -1, -1
6	5	2 2 3 2 4	3, 3, 4, 4, -1
7	1	10	-1

Sample Execution

```
Last login: Mon Oct 19 00:08:46 on ttys000
[bhavya@Bhavyas-MacBook-Air ~ % nano day51.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day51.c -o day51
[bhavya@Bhavyas-MacBook-Air ~ % ./day51
Enter the size of the array: 4
Enter 4 elements: 4 5 2 25
5, 25, 25, -1
bhavya@Bhavyas-MacBook-Air ~ % ./day51
Enter the size of the array: 4
[Enter 4 elements: 13 7 6 12
-1, 12, 12, -1
bhavya@Bhavyas-MacBook-Air ~ % ./day51
Enter the size of the array: 4
[Enter 4 elements: 1 3 2 4
3, 4, 4, -1
bhavya@Bhavyas-MacBook-Air ~ %
```